

Week 2 Task – Identifying Data Requirements

Primary Key

A **Primary Key (PK)** is a unique attribute that uniquely identifies each record in a database table. It cannot contain duplicate or NULL values. It helps maintain data integrity and establish relationships between different entities.

Entities, Attributes, and Primary Keys

1. Customer

Primary Key: Customer_ID

Attributes:

- Customer_ID (PK)
- First_Name
- Last_Name
- Email_ID
- Phone_Number
- Password
- Date_of_Birth
- Gender
- Address

2. Product

Primary Key: Product_ID

Attributes:

- Product_ID (PK)
- Product_Name
- Category_ID
- Supplier_ID
- Gender (*Male / Female / Unisex, if applicable*)
- Description
- Brand
- Unit_Price
- Stock_Quantity
- Weight
- Product_Image
- Manufacturing_Date
- Expiry_Date
- Product_Status

3. Category

Primary Key: Category_ID

Attributes:

- Category_ID (PK)
- Category_Name (*Examples: Electronics, Clothing, Footwear, Books, Beauty, Home Appliances, Sports, Grocery*)
- Category_Description

4. Supplier

Primary Key: Supplier_ID

Attributes:

- Supplier_ID (PK)
- Supplier_Name
- Contact_Person
- Email_ID
- Phone_Number
- Company_Name
- Address

5. Order

Primary Key: Order_ID

Attributes:

- Order_ID (PK)
- Customer_ID
- Order_Date
- Order_Status
- Total_Amount
- Shipping_Address
- Payment_Status
- Shipment_ID

6. Payment

Primary Key: Payment_ID

Attributes:

- Payment_ID (PK)
- Order_ID
- Customer_ID
- Payment_Date
- Payment_Method
- Payment_Status
- Transaction_ID
- Amount_Paid

7. Shipment

Primary Key: Shipment_ID

Attributes:

- Shipment_ID (PK)
- Order_ID
- Customer_ID
- Shipment_Date
- Delivery_Date
- Courier_Name
- Tracking_Number
- Shipping_Status
- Delivery_Address

8. Review

Primary Key: Review_ID

Attributes:

- Review_ID (PK)
- Customer_ID
- Product_ID
- Order_ID
- Rating
- Review_Comment
- Review_Date

Common Attributes Shared Across Entities

Some attributes are repeated in different entities because they establish relationships between tables.

Common Attribute	Appears In	Purpose
Customer_ID	Customer, Order, Payment, Shipment, Review	Identifies the customer associated with each record.
Product_ID	Product, Review	Identifies the product being reviewed.
Category_ID	Category, Product	Links each product to its category.
Supplier_ID	Supplier, Product	Links products with their suppliers.
Order_ID	Order, Payment, Shipment, Review	Connects orders with payments, shipments, and reviews.
Shipment_ID	Shipment, Order	Associates shipment details with an order.

Summary

The database consists of **eight main entities**: Customer, Product, Category, Supplier, Order, Payment, Shipment, and Review. Each entity has a **Primary Key (PK)** that uniquely identifies every record. Common attributes such as **Customer_ID, Product_ID, Category_ID, Supplier_ID, Order_ID, and Shipment_ID** are shared among different entities to create relationships between tables and maintain data consistency and referential integrity.

This version follows **database normalization** better and is the kind of structure lecturers usually expect for an e-commerce database assignment.